

Server Security and Automation

Theme: Startup Web Server Deployment

Section A: Scenario-Based Concept Application

1. **Explain the operational factors that justify selecting a specific Linux distribution for a startup's internal tool hosting environment.**

Answer:

- **Ubuntu Server** – Easy to use, large community, frequent updates.
- **CentOS Stream** – Stable, suitable for development/testing.
- **RHEL** – Enterprise support, high security, best for production.

Startup Recommendation:

- Ubuntu → Small startups.
- RHEL → Enterprise companies.
- CentOS Stream → Learning and testing.

2. **Explain the implementation of SSH Key-based Authentication to satisfy a security requirement where password logins must be disabled for developer access.**

Answer:

Generate SSH Key

```
ssh-keygen -t rsa -b 4096
```

Copy Public Key

```
ssh-copy-id user@server-ip
```

Disable Password Login

Edit SSH configuration:

```
sudo vim /etc/ssh/sshd_config
```

Change:

```
PasswordAuthentication no
```

```
PubkeyAuthentication yes
```

Restart SSH:

```
sudo systemctl restart sshd
```

Benefits

- Prevents brute-force attacks
- More secure than passwords
- Faster authentication

3. **Explain the configuration of Linux Permissions and Ownership for the `/var/www` directory to support a multi-user environment.**

Owner:

`sudo chown -R root:webadmins /var/www`

Permission:

`sudo chmod -R 775 /var/www`

Check:

`ls -ld /var/www`

Meaning:

775

Owner = rwx

Group = rwx

Others = r-x

4. **Explain how Logical Volume Management (LVM) facilitates dynamic storage expansion to prevent service downtime when disk space is depleted.**

Check disks

`lsblk`

Check Volume Groups

`vgdisplay`

Extend Logical Volume

`lvextend -L +5G /dev/vg/root`

Resize filesystem

`resize2fs /dev/vg/root`

Benefit:

- No downtime
- Dynamic storage
- Easy disk expansion

5. **Explain the troubleshooting process for diagnosing and resolving service blocks caused by SELinux security policies when a website becomes inaccessible.**

Check status

`getenforce`

View logs

`ausearch -m avc`

Temporary disable

`setenforce 0`

Permanent

`vim /etc/selinux/config`

Restart

`reboot`

6. **Explain the safe configuration of Cron Jobs to ensure critical daily backup tasks execute automatically at a scheduled time.**

Edit cron

```
crontab -e
```

Daily backup at 2 AM

```
0 2 * * * tar -czf /backup/web_$(date +%F).tar.gz /var/www
```

Check

```
crontab -l
```

Section B: Practical Application

Tools Required: Linux Distribution (Ubuntu/CentOS/RHEL), Terminal/CLI, Apache Web Server, UFW/Firewalld.

1. **System Identity: Set the system hostname to **startup-web-01** and verify the change via CLI.**

Check

```
hostname
```

Set

```
sudo hostnamectl set-hostname startup-web-01
```

Verify

```
hostnamectl
```

2. **Network Setup: Configure a static IP address on the primary interface and restart the networking service.**

Edit NetworkManager config

```
sudo nmcli connection modify ens160 \
```

```
ipv4.addresses 192.168.1.100/24 \
```

```
ipv4.gateway 192.168.1.1 \
```

```
ipv4.dns 8.8.8.8 \
```

```
ipv4.method manual
```

Restart

```
sudo systemctl restart NetworkManager
```

Verify

```
ip addr
```

3. **Secure Access: Enable the SSH service and ensure it starts automatically on system boot.**

Install:

```
sudo dnf install openssh-server
```

Start:

```
sudo systemctl start sshd
```

Enable:

```
sudo systemctl enable sshd
```

Verify:

```
systemctl status sshd
```

4. Access Control: Create a group named **webadmins** and a user **dev-ops** with restricted sudo privileges.

Create Group:

```
sudo groupadd webadmins
```

Create user:

```
sudo useradd dev-ops
```

Password:

```
sudo passwd dev-ops
```

Add to group:

```
sudo usermod -aG webadmins dev-ops
```

Restricted sudo:

```
sudo visudo
```

Add:

```
dev-ops ALL=(ALL) /bin/systemctl,/bin/journalctl
```

5. Web Hosting: Install the Apache service and deploy a custom **index.html** file to **/var/www/html**.

Install:

```
sudo dnf install httpd
```

Start:

```
sudo systemctl enable --now httpd
```

Create webpage:

```
echo "Welcome CloudScale" | sudo tee /var/www/html/index.html
```

Verify:

```
curl localhost
```

6. Hardening: Configure the system firewall to permit traffic only on ports 22 and 80.

Start firewall:

```
sudo systemctl enable --now firewalld
```

Allow SSH:

```
sudo firewall-cmd --permanent --add-service=ssh
```

Allow HTTP:

```
sudo firewall-cmd --permanent --add-service=http
```

Reload:

```
sudo firewall-cmd --reload
```

Verify:

```
firewall-cmd --list-all
```

7. Automation: Script a cronjob that executes at 02:00 daily to compress **/var/www** into a **.tar.gz** archive.

Edit:

```
crontab -e
```

Add:

```
0 2 * * * tar -czf /backup/www_$(date +%F).tar.gz /var/www
```

Verify:

```
crontab -l
```

Section C: Integrated Case Study

Scenario: "CloudScale," a boutique hosting provider, recently performed a kernel update on their primary web server. Upon reboot, the web application is unreachable (502 Gateway Error), and administrators report that SSH login attempts take over 60 seconds to respond. The server is live, but the business is losing revenue due to the downtime and unresponsive management interface.

Questions:

1. **Diagnosis:** Which specific log files and system commands would you use to identify why the Apache service failed to start?

Commands:

```
systemctl status httpd
```

```
journalctl -xe
```

```
journalctl -u httpd
```

```
apachectl configtest
```

Log files:

```
/var/log/httpd/error_log
```

```
/var/log/messages
```

2. **Connectivity:** Analyze why the SSH login is experiencing high latency and suggest a fix related to DNS or authentication settings.

Possible reasons

- Reverse DNS lookup
- Authentication timeout
- Slow DNS server

Fix:

```
sudo vim /etc/ssh/sshd_config
```

Add:

```
UseDNS no
```

Restart:

```
systemctl restart sshd
```

3. **Recovery: Outline the step-by-step CLI process to manually restart the failed services and verify their operational status.**

Restart Apache:

```
systemctl restart httpd
```

Restart SSH:

```
systemctl restart sshd
```

Check:

```
systemctl status httpd
```

```
systemctl status sshd
```

Verify ports:

```
ss -tuln
```

4. **Preventive Action: Propose a configuration change to ensure the web server automatically attempts to restart if it encounters a runtime failure.**

Edit service:

```
systemctl edit httpd
```

Add:

```
[Service]
```

```
Restart=always
```

```
RestartSec=5
```

Reload:

```
systemctl daemon-reload
```

Restart:

```
systemctl restart httpd
```

5. **Security: Justify the use of SSH Key-based authentication over passwords to prevent brute-force attacks during this vulnerable state.**

Advantages:

- Resistant to brute-force attacks
- No password guessing
- Strong cryptographic authentication
- Easier automation for servers
- More secure than passwords